

Vault Yield Tree — Reference

dxdt.fund • midas crawler • 2026-04-23

Goal

Reconstruct, per block, the full on-chain dependency graph of every tracked Morpho vault. Every node in the tree is a token or protocol the vault ultimately depends on for yield. Two use cases: risk tracing (“what actually backs this USDC deposit?”) and yield attribution (“which leaves generated the APY?”).

Pipeline

Ethereum RPC (event logs + multicall reads) → **midas crawler** (Rust, pm2 id 15) → **SQLite** (`midas.db`) → **midas-api** (Node/Express, :3001) → **Vercel rewrites** → **React frontend** with D3 tree.

Data model (3 tables)

<code>nodes</code>	one row per token/protocol ever discovered. Fields: <code>id</code> , <code>address</code> , <code>symbol</code> , <code>node_type</code> , <code>status</code> (<code>unresolved</code> / <code>unknown</code> / <code>resolved</code> / <code>final</code>), <code>final_reason</code> . Vault-invariant — a token is one row no matter how many vaults use it.
<code>token_deps</code>	canonical topology, interval semantics: (<code>parent_id</code> , <code>child_id</code> , <code>context_kind</code> , <code>block_opened</code> , <code>block_closed</code> , <code>circular</code>). A row opens when a dependency appears, closes when it disappears. Tree reconstruction = recursive CTE from the vault root.
<code>node_state</code>	one row per (<code>node.id</code> , <code>block</code>). Time-series of <code>share_price</code> , <code>total_assets</code> , <code>backed</code> , <code>oracle_price_usd</code> , etc. Populated every block via <code>ResolveSession</code> .

Node types and context kinds

node_type — what a node is:

`metamorpho` (v1 vault), `metamorpho_v2`, `lending_market` (Morpho Blue market), `erc4626` (other wrapper), `wrapped` (stETH → ETH), `stablecoin_synthetic` (USDe, USDS, USR, reUSD, rUSD), `offchain_attested` (RWA oracle), `leaf` / `native` (terminal: USDC, ~~WETH~~, ...), `circular_ref` (cycle placeholder).

context_kind — why an edge exists:

`asset` (parent holds child as yield-earning asset, the common case), `collateral` (Morpho market accepts child as collateral — *not* a yield path, risk models must ignore these), `backing` (multi-asset backing of a synthetic, e.g. USDe → {USDC, USDT, ...}), `cycle` (detected circular reference).

Per-block resolution: ResolveSession

Single `sync::catalog::ResolveSession` per block, shared across all vaults. Maintains an address-keyed cache. Each `resolve(addr)` call:

1. Cache hit → 0 RPC.
2. Cycle detected (address already in `in_progress`) → emit `circular_ref`.
3. DB fast-path: if `status = 'final'` → reconstruct from DB, 0 RPC.
4. Else run the resolver chain (`morpho_market`, `metamorpho`, `erc4626`, `ethena`, `sky`, `reservoir`, `re`, `leaf`), emit multicall RPC.

`commit()` runs a single SQLite transaction that upserts `nodes`, `node_state`, `token_deps`, and the legacy `edges` projection.

After the recent refactor, **bootstrap** reuses the same shared-session design as **follow mode**: one session per block for all vaults. Result: across 20 vaults, ~400 total market references collapse to 85 unique (4.75× dedup), and RPC per block drops from ~1200 naive to ~116.

Weights, yields, cycles

Weight of a leaf = share of the vault’s total assets flowing through it (market allocation multiplied recursively along the path). Stored per-block in the `edges` projection for the D3 view.

7-day yield per node = $(\text{share_price_now} - \text{share_price_7d_ago}) / \text{share_price_7d_ago}$, annualised where relevant.

Cycles (e.g. bbqUSDC contains steakUSDC which contains ...) are closed via `circular` rows in `token.deps` and rendered as dashed back-arrows in the tree.

Frontend (dxdt.fund)

- `/discover`: hybrid list of every tracked vault. One row = {name, asset, chain, TVL, 7-day APY, top-5 collateral stacked bar, markets count}. Data from `GET /api/vaults/summary` (cached 30s).
- `/vault/:slug` **Summary** tab: KPI row (TVL, 7d APY, markets, avg backed) + full collateral exposure bar + yields by market (filtered to this vault's own `lending_market` nodes).
- `/vault/:slug` **Dependency tree** tab: D3 hierarchical layout. Node radius $\propto \sqrt{weight}$. Colors by `node_type`. Dashed red arrows for cycles. Slider filters low-weight deps. Click a node to collapse/expand.

Invariants

- A market's `collateral` edge is *not* a yield path — risk analyses must skip that branch.
- `token.deps` is the canonical source of truth; `edges` is a projection kept for backward compat.
- Bootstrap and follow share the same writes; resuming a half-done run is safe via `sync_state.last_block`.
- KNOWN_LEAVES (USDC, USDT, PYUSD, USDtb, WETH, ETH, BTC sentinel) are pragmatic terminals: no resolver, always final.
- `/api/vaults/summary` $TVL = \text{root total.assets} \times \text{oracle-price.usd}$; for USD-pegged underlyings the oracle ≈ 1 .